Vector Database Architecture & Indexing Fundamentals

Section 1: HNSW vs IVF-PQ Vector Indexing Mechanics

Approximate Nearest Neighbor (ANN) search algorithms navigate a trade-off between recall accuracy, query latency, and index size. Hierarchical Navigable Small World (HNSW) is a graph-based index structure that provides state-of-the-art recall (>98%) and sub-millisecond query latency. HNSW builds a multi-layer graph where upper layers contain sparse long-distance skip links for rapid routing, and the bottom layer contains dense local connections. However, HNSW graphs require holding both vector vectors and graph edges in memory (RAM), requiring roughly 1.5 to 2.0 KB per 1536-dimensional vector. Conversely, Inverted File with Product Quantization (IVF-PQ) partitions the vector space into Voronoi cells and compresses vectors into compact byte codes. IVF-PQ reduces memory footprint by up to 95% compared to raw HNSW, but incurs an initial index training phase and suffers lower recall on out-of-distribution queries.

Section 2: LanceDB Columnar Architecture and Zero-Pod Economics

LanceDB is built on top of Lance, a modern columnar data format tailored for machine learning and multimodal data inspired by Apache Arrow and Parquet. Unlike traditional vector databases that run persistent container pods with gigabytes of allocated RAM (such as Pinecone Standard or Weaviate clusters), LanceDB is fundamentally serverless and disk-native. It executes vector distance calculations directly on NVMe SSDs or cloud object storage like AWS S3 or Google Cloud Storage. Lance organizes data into columnar batches with fast dictionary encoding, zero-copy reads via Arrow, and inverted vector indices stored directly inside the dataset files. Because there is no background daemon or always-on pod requirement, an idle LanceDB index incurs zero compute cost, making it the most cost-efficient choice for lightly-queried corpora.

Section 3: Qdrant Vector Engine Architecture and Payload Filtering

Qdrant is an open-source vector search engine written in Rust that specializes in real-time vector search with comprehensive payload filtering. Qdrant combines modified HNSW graphs with memory-mapped files (mmap) and on-disk payload storage. A key architectural advantage of Qdrant is its single-stage filtered search: instead of pre-filtering (filtering before vector search, which destroys HNSW graph connectivity) or post-filtering (filtering top-k results after vector search, which risks returning fewer than k items), Qdrant evaluates payload filters during the HNSW graph traversal. In embedded mode, Qdrant runs inside the application process without network latency, persisting index files and payloads to a local directory.

Section 4: Memory vs Disk Trade-offs across Vector Scales

At 100K vectors with 384 dimensions, raw vector data occupies approximately 150 MB. Both RAM-based and disk-based engines deliver sub-5ms retrieval. However, at 10M vectors, raw embeddings consume 15.36 GB, and HNSW indices expand the memory footprint to over 25 GB. On managed cloud providers, maintaining a 32GB RAM pod costs between $70 and $250 per month even if query traffic is zero. By contrast, storing 10M vectors on disk or S3 with LanceDB costs less than $0.50 per month in storage, with query latencies staying under 15ms on modern NVMe drives.